SWARM DRONE CHALLENGE

Regulations

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v4.0
Ingolstadt

Contents

Changelog

Current Version:
Version Draft v4.0

Previous Versions:
No previous Versions

Preface

Welcome to the Swarm Drone Challenge!

The Swarm Drone Challenge 2026 (SDC26) is an international competition dedicated to advancing the capabilities of autonomous aerial systems through the development and demonstration of swarm intelligence. It invites students, professionals, and startups to design and implement self-developed algorithms that enable groups of drones to cooperate effectively in complex, realistic mission environments.

Building on the success of previous editions, SDC26 continues the established Capture the Flag scenario, in which two teams deploy multiple drones operating collaboratively as a swarm. This competitive format promotes diverse technical and strategic approaches, addressing key challenges such as resilient communication, AI-based coordination, adaptive decision-making, and collective mission behavior under dynamic conditions.

The purpose of the Swarm Drone Challenge is to explore novel methods for distributed autonomy, evaluate innovative algorithmic concepts, and contribute to the evolution of common frameworks and standards for cooperative unmanned systems. By fostering experimentation and knowledge exchange among participants, SDC26 aims to identify promising technological directions and advance the understanding of swarm robotics in both theoretical and applied contexts.

Notes:

- **Contents marked in red are still subject to change and should be seen as current estimations and placeholders.**
- **This whole document is still a work in progress. Some points may still be left empty.**
- **If you have any comments on specific regulations, for example, think that they are not required, are too strict or don't make sense, don't hesitate to contact fly@brigkair.digital.**
- **You are alway invited to contribute to this document by creating a pull request on github.**

1 Competition Outlines

1.1 Playing Field

The playing field is a rectangular area with the following dimensions (Figure 2) :

- Length: 20 meters
- Width: 10 meters
- Height: 6 meters

The playing field is divided into three zones:

- Two Team Zones (red and blue): The home base of each team, where the drones start, land and targets are strategically located.
- One Neutral Zone in the middle: A neutral territory for strategic maneuvers.

The sizes shown in the graphic are approximate values. The exact dimensions may vary slightly.

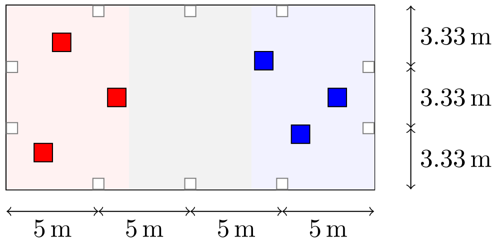

Figure 1: Field Dimensions

1.2 Target Objects

- The Goal: The objective of the game is to change the color of the target objects to your team's color (red or blue). All gameplay mechanics, scoring rules, and special conditions are described in detail in Section Gameplay.
- Each team zone has three target objects, for a total of 6 in the playing field.
- These objects change their color based on their interaction with the drones:
  - Default state: Matches the color of the team zone.
  - Triggered state: When an enemy drone approaches and targets the box, it changes to the attacking team's color, updates the ARUCO marker, and plays a horn signal. A subsequent enemy drone from the opposing team can then trigger it again.
- Dimensions of each target box:
  - Height: 55 cm (open: 73 cm), Width: 57.5 cm, Depth: 37.5 cm
- Detection: The target boxes detect approaching drones via an internal NFT tag attached to the drone. The system can reliably identify drones within a height range of approximately 1–2 meters. Further specifications and details will be provided in an upcoming update.

1.3 Gameplay

- Teams: 2–5 drones per team (independent team sizes), ready to battle!
- Start Setup: Drones may be placed inside their designated team zone and powered on within the cage.
- Cage Clearance: Once all human operators and staff have exited the cage and the area is confirmed clear, the cage is closed.
- Final Start Signal: After the cage is secured, the final start signal is given and all drones are permitted to take off and begin flight operations.

Figure 2: Example of an SDC target box.

- Action: Drones fly around the arena, try to trigger target objects to change color and score points for their team.
- Target Configuration: The positioning of the target objects within the arena will vary for each match. This ensures that teams cannot pre-program their drones for fixed setups and must rely on real-time perception, adaptability, and autonomous decision-making during gameplay.
- Scoring:
  - Scoring Methods:
    * Standard Capture (1 Point): Drone captures target box, holds for validation time, returns to home zone before re-capture.
    * Special Capture (5 Points): Single drone captures box, entire team returns to home base together.
    * Simultaneous Capture (10 Points): Multiple drones capture boxes at the same time.
  - Capture Stability: A target box must remain in the capturing team's color for at least **5 seconds** to validate capture. Own drone must hover over box for **2 seconds** to revert color change. Instant/rapid changes do not count.
  - Home Zone Validation: The drone or drone team must enter the home zone completely and remain inside for at least **5 seconds**. Brief or partial contact with the home zone does not validate a capture.
  - Simultaneous Capture: Bonus points are awarded only when two or more **different target boxes** are captured by different drones of the same team **simultaneously**.
  - Special Maneuver: When all six target objects display the same team color continuously for at least **5 seconds**, the team wins the game instantly. Temporary or unstable color states do not trigger a victory.
- Defense Strategy: Teams may defend target boxes by positioning drones nearby. Strictly prohibited: physical obstruction, continuous hovering over targets, capture system interference, and all forms of **jamming or electronic warfare**. **"Sitting dead duck" behavior** — that is, passive sitting or hovering directly above the boxes — is not allowed and will lead to **disqualification**.
- Time Limit: The game lasts for a maximum of 10 minutes. The team with the most points at the end wins the game!

1.4 Tournament

- Tournament Format: Follows classic elimination bracket with several rounds. Number of rounds depends on participating teams.
  - Each match played once per round.
  - Matches pit two teams head-to-head per rules in 1.3.
- Two teams face each other in each match.

- Pairings for the first round are determined by a random draw.
- The winner of each match is the team that scores the most points.
- Each team plays only once per round.
- If the number of teams is odd, one team will temporarily remain without an opponent:
  - Voluntary opponent teams are first requested.
  - If only one team volunteers, it is directly selected.
  - If multiple teams volunteer, the opponent is chosen by random draw.
  - If no teams volunteer, a random draw determines an opponent among all remaining teams.
- The team selected as an additional opponent will have two match results, and only the better result counts toward its ranking.
- In the event that an assigned opponent cannot participate, a replacement team is drawn by lot or, if unavailable, the match is scored as a technical win (default victory) for the other team.
- The final match is determining the overall tournament champion.

1.5 Winning the Challenge

To win the Swarm Drone Challenge, a jury of experts will evaluate the teams' performance in the game as well as their technical approach and the presentation of their work. The winning team is determined by the jury. The specific evaluation criteria will be noted and communicated at a later time.

1.6 Teams

- Up to 5 team members per team can be present on-site
- New team members are allowed to join but need to be registered with brigkAIR.

1.7 Reimbursement

- Teams are allowed to order parts and materials over time as needed, but must plan carefully since only five reimbursements in total will be processed.
- The combined value of all orders must not exceed €1000.
- Until the final, the hardware remains the property of brigkAIR and is provided to the teams as a permanent loan.

1.8 Safety and Fair Play Rules

- The teams are not allowed to start any motors before the official start signal is given by the judges.
- Before the match begins, each team must call "Ready" to the judges to confirm that all drones and operators are in a safe and prepared state.
- Entering the flight arena during the match is strictly prohibited for safety reasons.
- Any violation of these safety rules may result in disqualification from the current match or other disciplinary action as determined by the judges.
- Safety takes priority over scoring — teams must always operate their drones in a controlled and responsible manner.

1.9 Intellectual property

- The IP stays with the teams. All intellectual property rights of the hardware and software developed for this challenge remain with the respective teams.
- Exceptions: Any hardware, components, or materials provided by brigkAir remain the property of brigkAir and must be returned after the event if requested.

- Teams are encouraged to open-source their solutions. Doing so may have a positive influence on the jury's evaluation, as it promotes collaboration and knowledge sharing.
- Each team must submit a technical report of up to five (5) A4 pages, describing the design, implementation, and functionality of their solution.
- In addition, teams are required to prepare a presentation explaining their approach and key technical aspects to the judges during the evaluation phase.
- The presentation shall be five (5) minutes long, prepared in PowerPoint (.ppt) format, and held during both the Qualification and the Final rounds.

1.10 Responsibilities

- Each team must designate at least one Spotter, who assists the pilot in maintaining visual awareness of the drones and ensures safe operation within the arena.
- Each team must also appoint a Safety Officer, who serves as the primary contact for all safety-related matters between the team and the event committee.
  - The Safety Officer must be present during all matches.
  - They hold the authority to initiate an immediate stop of the match ("Killswitch") if any unsafe situation occurs.
  - The Safety Officer is responsible for ensuring that all team members follow the safety protocols at all times.
  - Prior to the Fly-Off, each team must submit information about their Safety Officer (name, role, and contact details) to the event organizers.
- Teams are responsible for virtually validating their solution before the competition to ensure functional and safe system behavior.

2 Technical Regulations

2.1 Positioning

The challenge location is still to be determined and will be announced during preparation phase. To improve the state estimation in a potential GPS denied environment, a camera based positioning system using ArUco marker can be used.

Two ArUco markers are attached to each pillar in the hangar. Figure 3 illustrates the IDs of the markers that are attached to the pillars, where the lower ArUco marker is mounted at a height of 2m and the upper marker at 4m. In the notation "X/Y", the first number (X) corresponds to the upper marker at 4m height and the second number (Y) to the lower marker at 2m height.

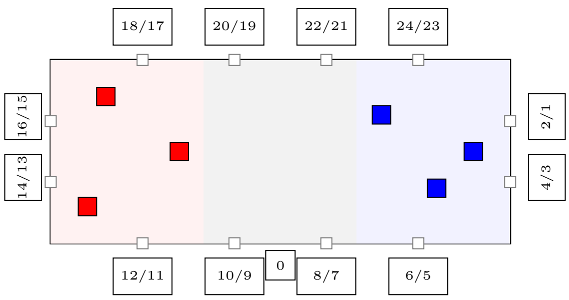

Figure 3: ArUco Marker ID's

None of the hangar's walls are aligned to the north or the east direction. To simplify the positioning system an additional marker with the ID 0 is placed at the back wall in the center of the field. This 0 marker can be used to attach the local world coordinate system to it.

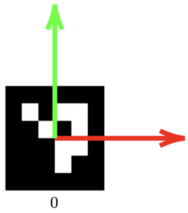

Figure 4: Origin point of the local coordinate frame

The ArUco markers are from the $DICT_{4\times450}$ dictionary and have a size of 0.5m x 0.5m. This information enables positioning in the arena with a view of a single marker.

2.2 Collision Prevention

- Each drone is recommended to be equipped with a spherical protection case, which helps to protect both the drone itself and other drones from damage during the match.
- The protective spheres must be provided by the teams themselves. brigkAir does not supply drone protection equipment.
- If a drone becomes damaged or inoperative during a match, no replacement drone may be added to the team's lineup. The team must continue the match with the remaining functional drones.
- Each team may register one (1) spare drone as a backup before the tournament begins. This spare drone can only be used to replace a malfunctioning unit between matches, not during an active match.
- Teams must always avoid intentional collisions or actions that could cause damage to other drones or the arena. Fair and safe flying behavior is mandatory.
- If a team's number of operational drones drops below two, the team will be disqualified from the current match for safety and fairness reasons.

2.3 Continuity of C2 Link

- A continuous Command and Control (C2) link between the operator and each drone must be maintained at all times.
- If the C2 link is interrupted or lost, an automatic flight termination or controlled landing must be initiated no later than 2 seconds after the loss of connection.
- In the event of a C2 link failure, the match will be paused or safely terminated by the judges. The affected team will not be penalized, as maintaining safety is the highest priority in this competition.
- Once the drone and the area are confirmed safe, the judges will decide whether the match can be resumed or restarted, depending on the situation and remaining time.

2.4 Emergency Shutoff and Kill Switch

- Each drone must be equipped with a physical or virtual kill switch, accessible to the operator or the team's Safety Officer, and its correct functionality must be demonstrated prior to the match.
- Activation of the kill switch must result in an immediate landing or motor shutdown of the drone.
- Each team must have one master button or command that can simultaneously land or shut down all drones of the team in case of an emergency.
- Improper or unsafe operation after a kill-switch event may lead to disqualification, but safe and responsible use of the emergency system will not result in penalties.

2.5 Size and weight limits

- The Maximum Take-Off Mass (MTOM) of each drone must be below 10 kg.The declared TOM includes all payload, batteries, and protective equipment.
- Each drone swarm may consist of a maximum of 5 drones and at least 2 drones.

2.6 Drone identification

- Each drone must be equipped with an RFID marker for automatic identification by the target objects.
- The RFID markers will be provided by brigkAIR at the event and must be properly attached according to the organizer's specifications.
- Teams are responsible for ensuring that the RFID marker is securely mounted and fully functional throughout the competition.

2.7 Swarming Functionality

For the purpose of this challenge, swarming functionality is defined as the coordinated and cooperative behavior of two or more drones that dynamically share information to achieve a common objective.

- Each team must implement and demonstrate a swarming functionality in accordance with the competition's definition of swarm behavior.
- The teams shall describe and explain their swarming approach in both the technical report and the presentation during the Qualification and Final stages.
- The demonstration must clearly show coordination and communication between multiple drones — i.e., the drones must act as a collective system rather than as independent units.
- The swarming functionality may be realized using different communication or sensing methods, for example:
  - Light-based signaling (e.g., LEDs, optical markers)
  - RF-based communication (e.g., Wi-Fi, LoRa, Bluetooth, UWB)
  - EO-based coordination (electro-optical vision systems, camera tracking)
  - Acoustic or infrared systems, or other innovative interaction methods

2.8 Ground Control Station

- Computing and control do not have to be performed exclusively onboard the drones. A combination of onboard processing and ground-based computation via a Ground Control Station is permitted.
- The swarm behavior may be governed by:
  - A single centralized agent that maintains awareness of the global swarm state, or
  - Distributed agents, where each drone operates semi-independently but coordinates with others through local communication or sensing.
- Teams must describe their chosen swarm architecture and communication topology in their technical documentation.

2.9 Arbiter / Judge

- A Ground Truth and Scoring Display visible to both teams is desired. This display should show the current match scores.
- However, no technical solution is provided by the organizers that collects or distributes real-time information on positioning, scoring, or any other referee-related data.
- Teams are therefore encouraged to develop their own sensing and evaluation systems if they wish to monitor swarm position or performance metrics during the match.